

Day 4 – Shell Scripting: Loops & Automation

Objective:

Learn for, while, and until loops to automate repetitive tasks in shell scripting.

1. For Loop:

Used when looping through a fixed list or known range.

Example: `for i in {1..5}; do echo $i; done`

Looping Through Arguments:

`for service in "$@"; do echo "Checking $service"; done`

Looping Through Files:

`FILES=(/var/log/*.log)`

`for file in "${FILES[@]}"; do echo $file; done`

2. While Loop:

Used when condition must remain true.

Example:

`COUNT=1`

`while [ "$COUNT" -le 5 ]; do`

`echo "Attempt $COUNT"`

`COUNT=$((COUNT + 1))`

`done`

Retry Logic Example (DevOps Style):

`COUNT=1`

`MAX=5`

`while [ "$COUNT" -le "$MAX" ]; do`

`if systemctl is-active --quiet nginx; then`

`echo "Service running"`

`exit 0`

`fi`

`sleep 2`

```
COUNT=$((COUNT + 1))
```

```
done
```

Best Practices:

- Always validate input arguments
- Use "\$@" for argument loops
- Use arrays for wildcard expansion
- Add sleep in retry loops
- Handle failure with exit codes

Day 4 completed successfully